AGE-RESTRICTED SALES

Alcohol Purchase Check

The compliance-first age check for alcohol sales.
Clears adults in a second — routes every borderline case to your staff.

A product proposal By Namrata Bhoyar

One question, repeated a thousand times a day

- German law sets two hard limits — 16+ for beer & wine, 18+ for spirits
- One wrong sale risks fines, licence suspension and reputational damage
- Manual ID checks are inconsistent, slow, and put staff in confrontations
- Skip a check to keep the queue moving, and the liability is yours

The cost of one mistake

Fines · per violation

Licence · suspension or loss

Brand · headline and trust damage

THE PRODUCT

One camera, one instant decision

Alcohol purchase check system reads a customer's face at the till, estimates whether they clear the legal age for what they're buying, and either clears the sale or hands it to your staff — in about a second.

Fast

Adults clear in about a second — no queue, no confrontation

Safe

Every borderline case routes to a human — never an automated guess on a minor

Compliant

Built to GDPR and the EU AI Act, with the paperwork to prove it

HOW THE MODEL WAS TRAINED

The data behind the model

UTKFACE — TRAINS THE MODEL

~23,700 face images with age, gender and race labels. Binned into 9 age groups (0–2 → 70+). This is the only data the model ever learns from.

UTKFace — Zhang, Song & Qi, Age Progression/Regression by Conditional Adversarial Autoencoder, CVPR 2017. University of Tennessee, Knoxville.
License: non-commercial research use only.

Obtained via Kaggle: UTKFace— Non-commercial research use only (official page confirmed).
Zhang, Z., Song, Y., & Qi, H. (2017). Age Progression/Regression by Conditional Adversarial Autoencoder.

THE MODEL, AND WHY

Why we chose MobileNetV2

1. **Runs on-device**

   Designed for phones and edge hardware — light enough to run at the till, so no face image ever leaves the store

2. **Transfer learning**

   Pre-trained on ImageNet, so it already understands faces — we fine-tune rather than train from scratch on limited data

3. **Deployment-realistic**

   A lightweight model is what a real point-of-sale device can actually run — not a research-only heavyweight

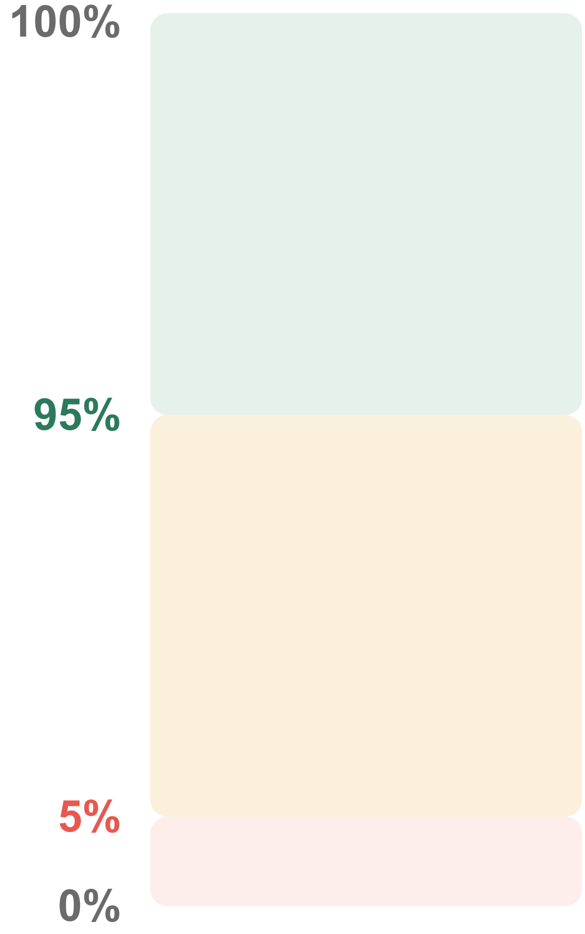
	Baseline	Final model
Input resolution	96px	**224px + augmentation**
Exact-band accuracy	46.6%	**54.7%**
Fairness gap (race)	18.5pp	**14.5pp**

Higher resolution plus augmentation improved accuracy and fairness together — the single biggest lever.

WHY IT'S SAFE TO BUY · OPTION A

How sure is 'sure enough'?

For every face, the model gives a confidence score — the chance the person is over the legal age. That score decides what happens next:



Over 95% sure → CLEAR Obviously an adult. Sale goes through instantly.

In between → CHECK ID Not sure enough. A staff member verifies. This is where every borderline teenager lands.

Under 5% → REFUSE Clearly under age. No sale.

The machine only acts alone on the certain cases. Every close call goes to a human — which is exactly the oversight the EU AI Act requires.

WHY IT'S SAFE TO BUY

Compliance isn't a feature — it's the product

GDPR

- ✓ No face image is ever stored
- ✓ Runs on-device — data never leaves the device
- ✓ Race & gender never used to make a decision
- ✓ Data-minimised and DPIA-ready by design
- ✓ Identifies no one — only estimates an age band

EU AI Act

- ✓ Not a prohibited practice
- ✓ Human oversight built into every borderline case
- ✓ Full technical documentation and model card included
- ✓ Performance measured and disclosed, not hidden
- ✓ Assessed against high-risk duties — ready for audit

REGULATORY ANALYSIS

The laws we checked ourselves against

GDPR

- Art. 5 — data minimisation & purpose limitation: race/gender only for auditing, never a decision input
- Art. 25 — protection by design: no face image stored, runs on-device
- Art. 35 — a DPIA is prepared before any real deployment
- The nuance: we identify no one, so this may not even be biometric data — we treat the strict rules as applying anyway

EU AI Act

- Not a prohibited practice — it infers no protected traits and identifies no one
- Art. 14 — human oversight is built into every borderline decision
- Art. 11/13 — full technical documentation and a model card ship with the product
- High-risk status is genuinely unresolved in the Act — so we assessed against those duties anyway, to be safe

What regulatory analysis means: *identify which laws apply, state exactly what they require, and check the project against each — before it ever reaches a real customer.*

WHY FAIRAGE, NOT THE ALTERNATIVE

We measure what others hide

TYPICAL AGE-ESTIMATION MODELS

- A black box — no idea why it decides what it decides
- One automated yes/no, even on borderline faces
- Accuracy claimed, never broken down by group
- Compliance left entirely to you

OUR MODEL

- ✓ Audited across demographic groups — gaps found and disclosed
- ✓ Borderline cases always routed to a human
- ✓ Every number measured, with confidence intervals
- ✓ GDPR & AI Act documentation ships with the product

The risks we found — and how we handle them

CRITICAL

A minor read as an adult

The 10–19 band straddles both legal limits and is the model's weakest zone.

→ borderline cases always go to a human

HIGH

Accuracy differs by race

Some groups sit in easier age ranges, so one metric alone can mislead.

→ report two fairness metrics; collect more data

HIGH

Proxy-feature bias

Race is never an input, but a model could rebuild it from pixels.

→ attention audited with Grad-CAM; risk kept open

PROOF — FULL TRANSPARENCY

The numbers, disclosed up front

90.3%

predictions within one
age band of the truth

< 5%

of the time is a sale
auto-cleared or auto-refused
without human review*

100%

of borderline cases
routed to your staff

We don't oversell the model. *Exact age-band accuracy is still improving, and we found a fairness gap across groups — which is exactly why the human-review layer exists. The system is designed to be safe even when the model is uncertain.*

*Everything in between goes to a human. Figures are from our current test set and disclosed in full in the model card.

WHAT YOUR INVESTMENT FUNDS

From pilot to full rollout

Now

Pilot

Deploy in a handful of stores; validate on your real camera feeds and lighting

3 mo

Harden

Multi-seed retraining, per-region legal thresholds, robustness testing

6 mo

Scale

Chain-wide rollout with live fairness monitoring and audit logging

Every stage tightens compliance and de-risks the next — you never scale ahead of the evidence.

THE ASK

Start with one pilot store

- Give us one location and a two-week pilot
- We deliver the model
- You measure the queue time saved and the checks caught — on your own floor
- If the numbers hold, we scale together

FairAge: *an age check your lawyers, your cashiers can all stand behind.*

THANK YOU